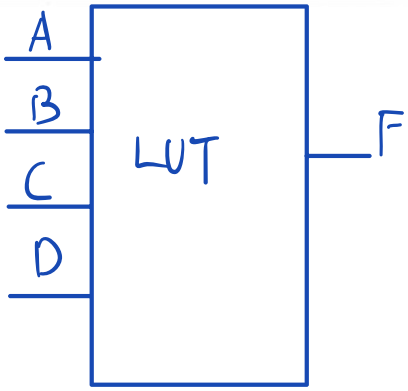
4.12 将一个四变量的 LUT 编程后产生如下信号：

$$F=\overline{A}BC\overline{D}+\overline{A}B\overline{C}\overline{D}+\overline{A}BCD+A\overline{B}C\overline{D}+AB\overline{C}D$$

解：



真值表：

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

4.13 Altera 公司 MAX7000 系列 CPLD 有什么特点？

答：采用 E^2PROM 技术编程，采用多电压 I/O 接口，I/O 既能和 5V 器件兼容，也能和 3.3V 器件兼容，有 6 个允许输出控制信号和 2 个全局时钟输入信号。

4.15 Xilinx 公司的 XC4000 器件主要由哪几部分组成？

答：① 可编程逻辑块 CLB

② 可配置存储器 SRAM 阵列

③ 可编程输入输出块

④ 可编程内部连线 PI

4.17 用 VHDL 设计 SOP 表达式 $X=AB+CD+EF$。

```
library IEEE;
use IEEE.std_logic_1164.all
entity SOP is
   port (A,B,C,D,E,F: in std_logic;
         X: out std_logic);
end SOP;
architecture behave of SOP is
begin
   X<= (A and B) or (C and D) or (E and F);
end behave;
```

4.18 用 VHDL 设计布尔表达式 $F=(A+\overline{B}+C)(A+B+\overline{C})(\overline{A}+\overline{B}+\overline{C})$。

```
library IEEE;
use IEEE.std_logic_1164.all
entity bool is
   port (A,B,C: in std_logic;
         F: out std_logic);
end bool;
architecture behave of bool is
begin
   F<= (A or (not B) or C) and (A or B or (not C) and ((not A) or (not B) or (not C));
end behave;
```

4.19 用 VHDL 设计 3 线-8 线译码器。

```
library IEEE;
use IEEE.std_logic_1164.all
entity decoder is
    port (A,B,C,G1,G2A,G2B in std_logic;
          Y: out std_logic_vector (7 downto 0));
end decoder;
architecture behave of decoder is
    signal X: std_logic_vector (2 downto 0);
    begin
        X = C&B&A;
        process (X, G1, G2A, G2B) begin
            if (G1='1' and G2A='0' and G2B='0') then
                case X is
                    when "000" => y <= "11111110";
                    when "001" => y <= "11111101";
                    when "010" => y <= "11111011";
                    when "011" => y <= "11110111";
                    when "100" => y <= "11101111";
                    when "101" => y <= "11011111";
                    when "110" => y <= "10111111";
                    when "111" => y <= "01111111";
```

```
            end case;
        else y <= "11111111";
        end if;
    end process;
end behave;
```

4.20 用 VHDL 设计一个 3 位 Gray 码可逆计数器，y=1 时计数器实现“加”运算，y=0 时计数器实现“减”运算。

```
library IEEE;
use IEEE.std_logic_1164.all
entity counter is
    port ( y, clk : in std_logic;
           Q : out std_logic_vector (2 downto 0));
end counter;
architecture behave of counter is
    signal d: std_logic_vector (2 downto 0);
    begin
        process (clk) begin
            if (clk'event and clk = '1') then
                if (y = '1') then
                    case d is
                        when "000" => d <= "001";
                        when "001" => d <= "011";
```

```
            when "011" => d <= "010";
            when "010" => d <= "110";
            when "110" => d <= "111";
            when "111" => d <= "101";
            when "101" => d <= "100";
            when "100" => d <= "000";
            when others => d <= "000";
        end case;
    end if;
    if (y='0') then
        case d is
            when "000" => d <= "100";
            when "001" => d <= "000";
            when "011" => d <= "001";
            when "010" => d <= "011";
            when "110" => d <= "010";
            when "111" => d <= "110";
            when "101" => d <= "111";
            when "100" => d <= "101";
            when others => d <= "000";
        end case;
    end if;
```

```
        end if;
    end process;
    Q <= d;
end behave;
```